

Adversarial Machine Learning CIA-3

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Attack and Defense Techniques

Demonstration using the Breast Cancer Wisconsin Diagnostic Dataset

Machine Learning Security – Attack and Defense Analysis

1. Objective

The aim of this experiment was to study how machine-learning models behave when they are exposed to adversarial attacks. The experiment focused on two situations: decision-time evasion attacks, where the input is changed after the model has been trained, and data-poisoning attacks, where the training data is changed before training. The Breast Cancer Wisconsin Diagnostic dataset was used throughout the experiment. Four ML classifiers, two clustering techniques, a linear model and a deep-learning model were used. Different defense techniques were then tested to see how suspicious or uncertain predictions could be handled.

2. Dataset Description

The open-source Breast Cancer Wisconsin Diagnostic dataset was used for the experiment. It contains 569 observations and 30 numerical features describing characteristics of breast cell nuclei. The target has two classes: malignant and benign. There are 212 malignant samples and 357 benign samples. A stratified split was used, giving 426 training samples and 143 testing samples. Feature scaling was applied before model training.

The dataset is suitable for this experiment because the original classification problem can be solved with high accuracy, which gives a useful baseline for studying how much the predictions change when the input or training data is manipulated.

3. Experimental Setup

Four classifiers were trained on the breast-cancer dataset: Logistic Regression, Decision Tree, Random Forest and Support Vector Machine (SVM). Logistic Regression was used as the required linear model. K-Means and DBSCAN were used for clustering and novelty detection. A PyTorch multilayer perceptron (MLP) was used for the deep-learning poisoning experiments.

4. Baseline Classification Results

Before introducing attacks, the four classifiers were evaluated on the original test data. These results provide the clean-data baseline.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	98.60%	98.89%	98.89%	98.89%	99.77%
SVM	97.90%	98.88%	97.78%	98.32%	99.69%
Random Forest	95.80%	95.65%	97.78%	96.70%	99.50%
Decision Tree	93.71%	95.51%	94.44%	94.97%	91.86%

The Logistic Regression model gave the best overall accuracy on this dataset (98.60%), followed by SVM (97.90%). This shows that the 30 numerical features contain enough information to separate the malignant and benign classes very well. These clean results are important because they give us something to compare against after introducing attacks.

5. Decision-Time Evasion Attacks

For the decision-time attacks, the models were first trained normally. The attack was then applied to test inputs by changing their feature values within the selected attack settings. The purpose was to see whether a small or controlled change to a breast-cancer sample could make the classifier produce a different class prediction.

Twelve decision-time attack attempts were carried out across the four classifiers. Ten of the twelve attacks were successful, giving an observed evasion success rate of 83.33%.

Classifier	Successful attacks	Result
Logistic Regression	3/3	All attempts successful
SVM	3/3	All attempts successful
Random Forest	2/3	One attempt resisted
Decision Tree	2/3	One attempt resisted

For this breast-cancer dataset, the result shows that good clean accuracy does not automatically mean that the classifier is secure. Logistic Regression and SVM were affected in all three of their tested attacks. Random Forest and Decision Tree resisted one attack each. The two unsuccessful attacks are also useful because they show that an evasion attack does not always work and that its success depends on the model and the selected perturbation.

6. Probability-Based Decision Analysis

Probability was used to show how a model's confidence can be included in the decision process. The normal binary threshold was 0.50, while a stricter 0.80 threshold was used as a defense policy.

Probability for Class 1	Threshold 0.50	Threshold 0.80
0.95	Class 1	Class 1
0.75	Class 1	ABSTAIN
0.51	Class 1	ABSTAIN
0.48	Class 0	ABSTAIN

This shows why confidence can be useful for defense. At the usual 0.50 threshold, even a probability of 0.51 is enough to make a Class 1 decision. With the stricter 0.80 threshold, the same case is treated

as uncertain instead. One actual evaluated case had a probability of 0.5594, so the defense did not automatically accept the prediction. This is useful in a breast-cancer application because uncertain cases can be referred for additional checking rather than being forced into a class.

7. K-Means and DBSCAN for Decision-Time Defense

K-Means and DBSCAN were used to check whether test inputs looked unusual compared with the breast-cancer training data. K-Means used a 95th-percentile distance threshold of approximately 7.9703. DBSCAN used the density of the training data to identify core and novel points.

Technique	Observed result
K-Means	0 novelty flags
DBSCAN	1 novelty flag

K-Means did not flag any of the evaluated test samples, whereas DBSCAN identified one novel sample. This difference is reasonable because K-Means mainly looks at distance from cluster centres, while DBSCAN looks at local density. Therefore, using both methods gives two different views of whether a breast-cancer sample looks unusual.

8. Decision-Time Defense Strategy

The decision-time defense was designed as a layered process rather than depending on a single check. Input range validation and feature clipping were used to control unrealistic values. Probability thresholding and low-confidence abstention were used to avoid accepting uncertain predictions. K-Means and DBSCAN were used as novelty checks, while disagreement between the four classifiers was treated as another warning signal.

For example, when a prediction did not reach the required confidence, the system returned ABSTAIN instead of directly accepting the predicted class. Similarly, a sample identified as novel by DBSCAN was rejected for automatic decision-making. This approach is more cautious than simply trusting the classifier output.

9. Deep Learning and Data-Poisoning Attacks

The same breast-cancer dataset was then used for the poisoning experiment. A PyTorch multilayer perceptron was trained as the deep-learning classifier. The clean MLP achieved 97.20% accuracy and 97.75% F1-score.

Eight poisoning scenarios were created. They included random label flipping, boundary-targeted label flipping and feature outlier injection. For the main defense comparison, 10% outlier poisoning was used. The poisoned model's accuracy fell from 97.20% to 94.41%, while its F1-score fell from 97.75% to 95.65%.

This result shows that the training data itself can be a security problem. In this case, the test samples were not changed; instead, the model learned from modified breast-cancer training observations, which affected its later predictions.

10. Poisoning Defense Results

Training condition	Accuracy	F1-score	Samples flagged
Clean MLP	97.20%	97.75%	0
10% outlier poisoned	94.41%	95.65%	0
Conservative K-Means + DBSCAN	97.90%	98.32%	54
Aggressive K-Means or DBSCAN	95.10%	96.09%	148

The conservative strategy required both K-Means and DBSCAN to flag a sample before removing it. It flagged 54 samples and improved the poisoned model's accuracy from 94.41% to 97.90%. This was the strongest poisoning-defense result in the experiment. The aggressive strategy removed samples flagged by either method. It flagged 148 samples but achieved only 95.10% accuracy.

This result is important for this dataset because breast-cancer observations can naturally differ from one another. Removing too many unusual observations can also remove legitimate data. The comparison therefore shows the trade-off between detecting more suspicious samples and keeping useful training data.

L2 regularization with early stopping was also tested. In this particular poisoning experiment, F1 decreased from 95.65% to 94.57%. Therefore, it did not improve the poisoned model in this setting. This is useful evidence that a general overfitting control method is not necessarily an effective defense against every poisoning attack.

11. Evidence for the 20 Defense Strategies

Twenty defense strategies were demonstrated or evaluated. The evidence below summarizes the main outputs obtained from the notebook.

No.	Defense	Evidence from experiment	Inference
D01	Input Validation Range	0 range violations	Tested inputs stayed within the expected feature ranges.
D02	Feature Clipping	0 post-clipping violations	Clipping successfully kept values inside the selected bounds.
D03	Consistent Scaling	30-dimensional vector	The transformed breast-cancer input retained the expected 30 features.
D04	Probability Thresholding	Best Logistic threshold = 0.50	The conventional threshold was used as the basic decision rule.
D05	Low-Confidence Abstention	9 cases at 0.80 policy	Uncertain predictions can be stopped instead of being automatically accepted.
D06	K-Means Novelty Detection	0 flags	No evaluated sample crossed the K-Means novelty threshold.

D07	DBSCAN Novelty Detection	1 flag	One evaluated sample was considered novel by density-based detection.
D08	Ensemble Agreement	80% agreement	Model disagreement can be used as an uncertainty signal.
D09	Perturbation Consistency	1.0 consistency	The evaluated perturbation check produced stable predictions.
D10	Human Review	Review ABSTAIN/disagreement cases	Uncertain cases can be passed to a human instead of forcing an automatic decision.
D11	Data Provenance	SHA-256 hash generated	The dataset can be checked for later modification.
D12	Duplicate Removal	0 duplicate rows	No duplicate training observations were found.
D13	Robust Scaling	Median 0, IQR 1	The robust-scaling transformation behaved as expected.
D14	Statistical Outlier Filtering	56 candidates	Some training observations were identified for further outlier checking.
D15	K-Means Screening	13 candidates	K-Means identified potential suspicious training observations.
D16	DBSCAN Screening	117 noise points	DBSCAN was more aggressive in identifying low-density training points.
D17	Label Consistency	97.89% agreement	Most labels agreed with out-of-fold model predictions.
D18	Drift Monitoring	0.057334 drift score	The measured train-test mean drift was relatively small.
D19	Regularization + Early Stopping	F1: 95.65% → 94.57%	This defense did not improve the tested poisoned model.
D20	Clean Validation Holdout	97.20% accuracy	The clean holdout provides a reference for measuring poisoning effects.

These results should not be interpreted as meaning that all 20 defenses stopped an attack. Some defenses are detection, preprocessing, monitoring or decision policies. The important point is that each strategy was demonstrated with a corresponding result or evidence entry.

12. Overall Discussion

Looking at the experiment as a whole, the Breast Cancer Wisconsin dataset gave very good clean classification results, but the attack experiments showed that accuracy alone is not enough to judge model security. Ten of the twelve tested decision-time attacks were successful. The models therefore need additional checks when they receive inputs that may have been manipulated.

The probability-based defense was useful because it allowed the system to treat uncertain predictions differently from confident ones. K-Means and DBSCAN provided another layer by checking whether an input looked unusual compared with the training data. Ensemble disagreement added another warning when the four classifiers did not make the same decision.

The poisoning experiment showed a similar issue during training. The 10% outlier poisoning reduced the MLP accuracy from 97.20% to 94.41%. Conservative K-Means and DBSCAN screening recovered the performance to 97.90%, while aggressive screening was less effective. This suggests that a balanced defense is preferable to simply removing as many suspicious samples as possible.

13. Limitations

The experiment was performed on one binary classification dataset, so the results should not be assumed to apply directly to every machine-learning problem. Attack success depends on the chosen perturbation settings, target samples and classifier. Similarly, K-Means and DBSCAN results depend on their thresholds and parameters. An outlier is also not automatically a poisoned sample because legitimate breast-cancer observations can naturally be different from the majority.

14. Conclusion

The experiment successfully demonstrated both decision-time evasion and data poisoning using the Breast Cancer Wisconsin Diagnostic dataset. The original classifiers achieved high accuracy, but 10 out of 12 decision-time attacks were able to change predictions. The deep-learning model was also affected when poisoned training data was introduced.

Among the tested defenses, the combination of K-Means and DBSCAN with conservative screening gave the best recovery in the poisoning experiment. Probability-based abstention, novelty detection and ensemble disagreement also provided useful decision-time safeguards. Overall, the experiment shows that protecting a model requires more than achieving high accuracy on clean data. Combining several defenses gives a more practical approach to handling adversarial inputs and suspicious training data.

15. Experimental Workflow

Breast Cancer Wisconsin Dataset → Preprocessing and Scaling → Four ML Classifiers → Clean Baseline Results → Decision-Time Evasion Attacks → Probability / K-Means / DBSCAN / Ensemble Defenses

Breast Cancer Training Data → Poisoning Attacks → PyTorch MLP → Performance Drop → K-Means + DBSCAN Screening → Defended Model